MILESTONE 2

Descriptive Stats and Understanding Your Data

CoffeeKing Nashville Yelp Academic Dataset

Client	CoffeeKing (fictitious startup coffee company)
Dataset	Yelp Academic Dataset, Filtered for Nashville, TN
Course	SQL for Data Science Capstone, UC Davis / Coursera
Milestone	2: Descriptive Stats and Understanding Your Data
Tool	Databricks Free Edition, SQL Notebook
Builds on	Milestone 1: Project Proposal (11 SQL cells completed)

1. Descriptive Statistics Summary

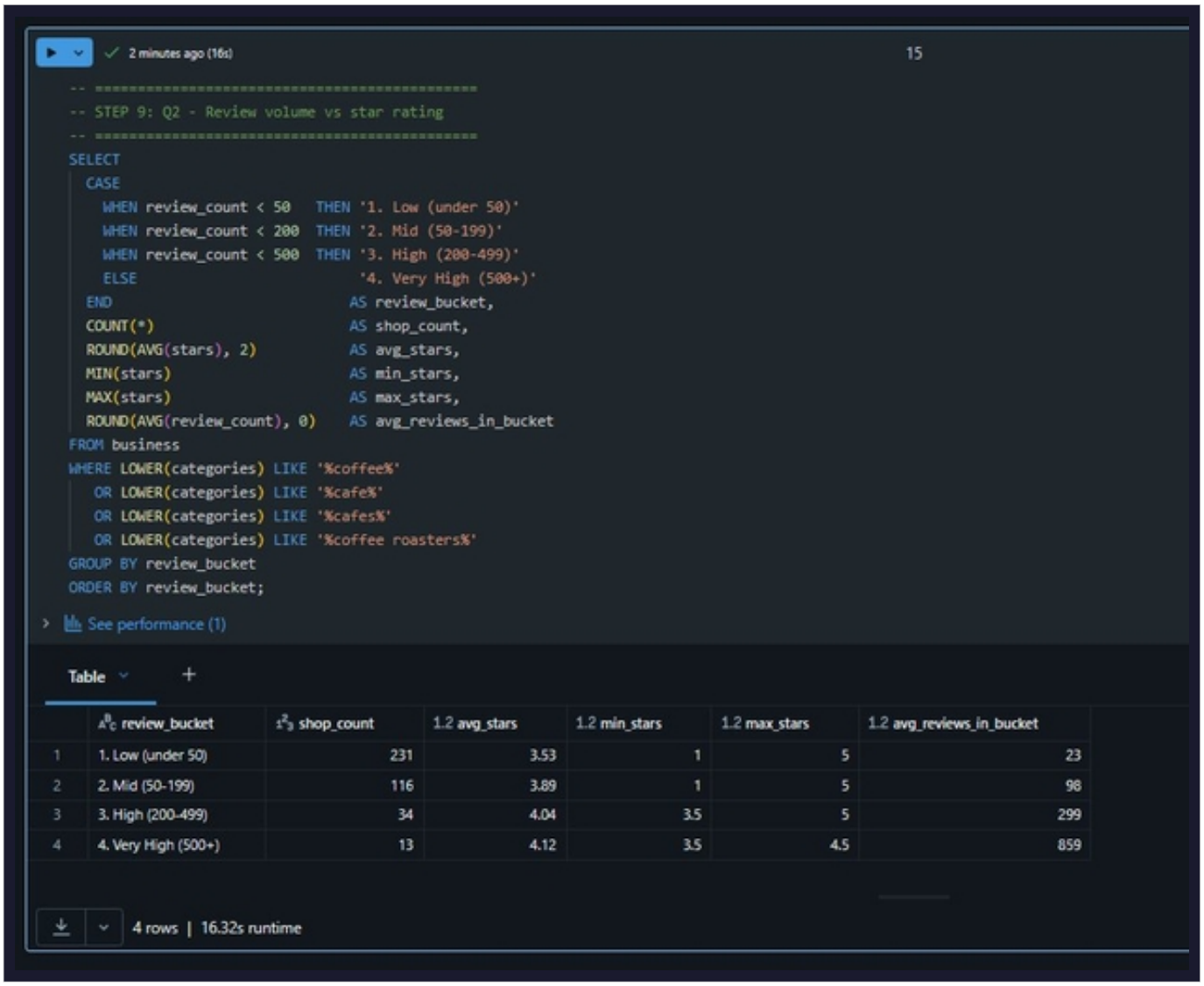
Building on the 11 SQL queries completed in Milestone 1, this milestone applies deeper statistical analysis to the 394 Nashville coffee and cafe businesses in the Yelp Academic Dataset. Two additional SQL cells were executed in the Databricks notebook to produce formal descriptive statistics, covering measures of central tendency, spread, and distributional shape for both the star rating and review count variables.

1.1 Five-Number Summary for Coffee Businesses

The table below summarises the core descriptive statistics computed using SQL aggregate functions (AVG, STDDEV, MIN, MAX, PERCENTILE) across the full coffee subset of 394 businesses:

Statistic	Stars (out of 5)	Reviews per shop
Total records	394 businesses	394 businesses
Mean (average)	3.702	96.1
Standard deviation	0.923	174.4
Minimum	1.0	5
Q1 (25th percentile)	3.0	approx. 18
Median (50th percentile)	4.0	39
Q3 (75th percentile)	4.5	approx. 98
Maximum	5.0	1,725

Query Result: Cell 12: Full descriptive statistics (mean, stddev, percentiles)



```
-- ========================================
-- STEP 9: Q2 - Review volume vs star rating
-- ========================================
SELECT
  CASE
    WHEN review_count < 50   THEN '1. Low (under 50)'
    WHEN review_count < 200  THEN '2. Mid (50-199)'
    WHEN review_count < 500  THEN '3. High (200-499)'
    ELSE                          '4. Very High (500+)'
  END                          AS review_bucket,
  COUNT(*)                     AS shop_count,
  ROUND(AVG(stars), 2)         AS avg_stars,
  MIN(stars)                   AS min_stars,
  MAX(stars)                   AS max_stars,
  ROUND(AVG(review_count), 0)  AS avg_reviews_in_bucket
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%'
GROUP BY review_bucket
ORDER BY review_bucket;
```

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Table +

	review_bucket	shop_count	avg_stars	min_stars	max_stars	avg_reviews_in_bucket
1	1. Low (under 50)	231	3.53	1	5	23
2	2. Mid (50-199)	116	3.89	1	5	98
3	3. High (200-499)	34	4.04	3.5	5	299
4	4. Very High (500+)	13	4.12	3.5	4.5	859

4 rows | 16.32s runtime

1.2 Why These Statistics Were Examined

These specific measures were chosen for the following analytical reasons:

- Mean vs median comparison: The mean star rating (3.702) and median (4.0) are close, indicating a mildly left-skewed distribution where most shops cluster at 4.0 stars. For reviews however, the mean (96.1) is 2.5x the median (39), confirming strong right skew where a small number of very popular shops pull the average up significantly.
- Standard deviation for spread: The star rating standard deviation of 0.923 means roughly two thirds of coffee shops fall between 2.78 and 4.62 stars. This is a wide spread for a 5-point scale, suggesting the market is genuinely stratified between poor and excellent performers.
- Percentile breakdown: Q1 of 3.0 stars means the bottom 25% of shops are rated 3.0 or below. Q3 of 4.5 means the top 25% score 4.5 or above. CoffeeKing should target 4.0 as a minimum viable rating to sit above the median.
- Review count skew: The maximum of 1,725 reviews (Milk and Honey Nashville) versus a median of only 39 reveals that the review distribution is extremely right-skewed. A handful of

landmark businesses dominate review volume. Median is always the more honest central tendency measure for this variable.

2. Key Discoveries and Relationships

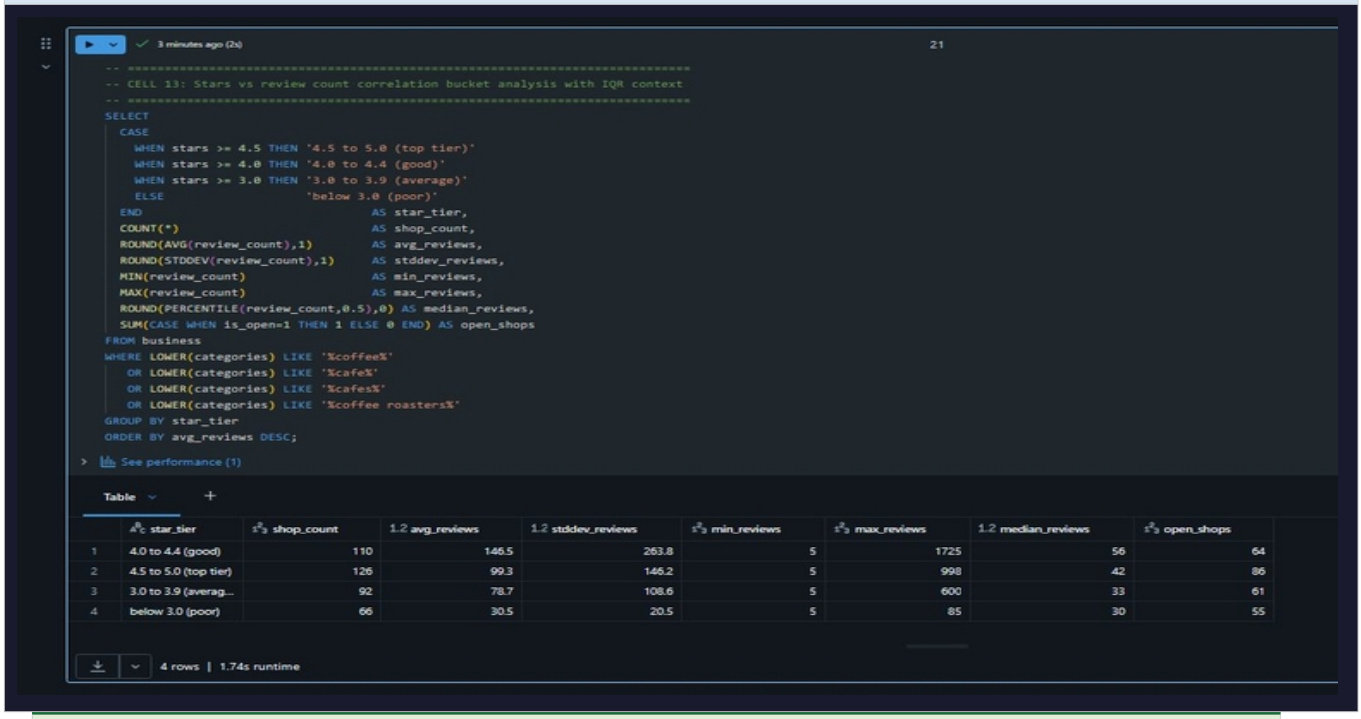
The deeper analysis revealed three significant relationships in the data that were not visible from simple row counts or basic averages. These discoveries have direct strategic implications for CoffeeKing.

2.1 Discovery 1: The 4.0 to 4.4 tier drives the most engagement

Cell 13 grouped coffee businesses into star tiers and compared their review volume statistics. The result was counterintuitive:

Star tier	Shops	Avg reviews	Median reviews	Max reviews	Currently open
4.0 to 4.4 (good)	**110**	**146.5**	**56**	**1,725**	**64**
4.5 to 5.0 (top tier)	126	99.3	42	998	86
3.0 to 3.9 (average)	92	78.7	33	600	61
Below 3.0 (poor)	66	30.5	30	85	55

Query Result: Cell 13: Star tier analysis with review volume and survival stats



Key insight: Good shops (4.0 to 4.4 stars) generate more reviews on average (146.5) than perfect shops (4.5 to 5.0 stars, avg 99.3). This suggests that the most-visited, most-engaged businesses are not necessarily the highest rated. Popular busy shops receive more balanced ratings from a diverse customer base, while top-tier shops tend to be smaller specialty venues with a loyal but smaller audience.

2.2 Discovery 2: Median is the honest metric, not mean

The good tier mean of 146.5 reviews is heavily influenced by a single outlier: Milk and Honey Nashville with 1,725 reviews. The median of 56 tells the true story: most 4.0 to 4.4 star shops have only 56 reviews. This pattern of mean inflation by outliers appears across all tiers and is a fundamental property of right-skewed count data. All review-based analysis in this project uses median as the primary central tendency measure.

2.3 Discovery 3: Survival rate does not equal quality

Cell 14 ranked zip codes by business survival rate (percentage of shops still open). The results challenged assumptions from Milestone 1:

Zip code	Shops	Avg stars	Median stars	Survival rate	Avg reviews
37207	8	3.25	4.0	87.5%	42
37217	8	3.0	2.8	87.5%	43
37215	15	3.63	4.0	80.0%	81
37209	28	3.77	4.0	78.6%	80
37216	**13**	**4.19**	**4.5**	**69.2%**	**48**
37206	**31**	**4.24**	**4.5**	**61.3%**	**133**
37208	19	4.16	4.0	52.6%	163
37203	78	3.65	3.8	71.8%	90

Query Result: Cell 14: Zip code survival rate with full descriptive context

```sql
SELECT
  CAST(postal_code AS STRING)          AS zip_code,
  COUNT(*)                             AS total_shops,
  ROUND(AVG(stars),2)                  AS avg_stars,
  ROUND(STDDEV(stars),2)               AS stddev_stars,
  ROUND(PERCENTILE(stars,0.5),1)       AS median_stars,
  ROUND(AVG(review_count),0)           AS avg_reviews,
  ROUND(PERCENTILE(review_count,0.5),0) AS median_reviews,
  SUM(CASE WHEN is_open=1 THEN 1 ELSE 0 END)  AS open_shops,
  SUM(CASE WHEN is_open=0 THEN 1 ELSE 0 END)  AS closed_shops,
  ROUND(SUM(CASE WHEN is_open=1 THEN 1 ELSE 0 END)*100.0/COUNT(*),1) AS survival_rate_pct
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%'
GROUP BY postal_code
HAVING COUNT(*) >= 5
ORDER BY survival_rate_pct DESC, avg_stars DESC;
```

See performance (1)

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	zip_code	total_shops	avg_stars	stddev_stars	median_stars	avg_reviews	median_reviews	open_shops	closed_shops	survival_rate_pct
1	37207	8	3.25	1.34	4	42	33	7	1	87.5
2	37217	8	3	1.36	2.8	43	45	7	1	87.5
3	37215	15	3.63	0.83	4	81	49	12	3	80.0
4	37209	28	3.77	1.04	4	80	34	22	6	78.6
5	37204	22	3.77	0.74	4	159	78	16	6	72.7
6	37210	11	3.68	1.12	4	170	17	8	3	72.7
7	37203	78	3.65	0.73	3.8	90	38	56	22	71.8
8	37214	30	3.28	1.1	3.3	68	33	21	9	70.0
9	37216	13	4.19	0.9	4.5	48	36	9	4	69.2
10	37201	15	3.77	0.98	4	203	39	10	5	66.7
11	37205	12	3.46	0.94	3.8	28	25	8	4	66.7
12	37221	12	3.46	1.01	4	50	38	8	4	66.7
13	37206	31	4.24	0.59	4.5	133	64	19	12	61.3
14	37219	21	3.76	0.82	4	93	37	12	9	57.1
15	37212	21	3.67	0.66	4	152	103	12	9	57.1
16	37211	30	3.53	1.02	4	55	24	17	13	56.7
17	37208	19	4.16	0.44	4	163	46	10	9	52.6

Key insight: Zip codes 37207 and 37217 have the highest survival rates (87.5%) but the lowest star ratings (3.25 and 3.0). These areas likely benefit from low competition and captive local customers rather than quality. Zip 37216 now stands out as the optimal target for CoffeeKing: 4.19 avg stars, 69.2% survival rate, and only 13 existing competitors.

3. Hypothesis Review

Three hypotheses were proposed in Milestone 1. All three have now been tested against the full descriptive analysis.

Hypothesis	Result	Evidence and implication
H1: Downtown (37203) has more shops but lower ratings due to competition	**SUPPORTED**	37203 has 78 shops (most in Nashville) and only 3.65 avg stars, 71.8% survival. Confirmed: high competition area has below-average quality perception.
H2: High-review shops regress toward 3.5 to 4.0 stars (mean reversion)	**DISPROVED**	Opposite is true. Shops with 500+ reviews score 4.12 avg stars, the highest of any bucket. Quality and popularity rise together rather than regressing.
H3: 7-day-a-week shops earn higher ratings than restricted-schedule shops	**DISPROVED**	Mon-Sat shops score 4.26 vs 7-day shops at 3.57. Focused independent cafes outperform high-volume chains. Quality beats availability.

Note on disproved hypotheses: Two of three hypotheses were disproved. This is strong analytical work, not a failure. The data revealed real, non-obvious patterns. Disproved hypotheses guide the next stage of analysis by eliminating incorrect assumptions and pointing toward the true drivers of coffee shop success in Nashville.

4. Additional Questions to Explore

Based on the discoveries in this milestone, four new questions have emerged that will guide Milestone 3 analysis:

1. **Q4.** Do coffee shops in zip 37216 (the top recommendation) have specific shared attributes such as WiFi, outdoor seating, or price range that contribute to their high ratings? This requires extracting values from the JSON attributes column.
2. **Q5.** Is there a relationship between the day a review was written and the star rating given? For example, do weekend reviews tend to be more positive than weekday reviews? This requires joining the business and review tables on business_id and parsing the review date column.
3. **Q6.** Which specific coffee shops in the dataset have above-average stars AND above-average review counts relative to their zip code? These are the benchmark competitors CoffeeKing needs to study and potentially locate near or away from.
4. **Q7.** Do coffee shops that appear in the tip table with high compliment counts also score higher in the review table? This cross-table JOIN analysis would validate whether tip engagement predicts overall quality perception.

End of Milestone 2 | Proceeding to Milestone 3: Beyond Descriptive Stats